

Supplementary Materials:
A multiscale covariance-aware statistical test for clustering in
single-molecule localization microscopy

Table S1: Full FPR Calibration

Table 1: Complete false positive rate calibration from 200 CSR datasets with 50 mocks each. Combined FPR uses the OR rule ($p_{\text{var}} < \alpha$ or $p_{\text{skew}} < \alpha$). Per-test FPR at $\alpha = 0.05$: variance = 9.0% (18/200), skewness = 9.0% (18/200). Mean p -values: $\bar{p}_{\text{var}} = 0.464$, $\bar{p}_{\text{skew}} = 0.463$.

α	Combined FPR	n detected	95% CI lower	95% CI upper	DBSCAN FPR	Ripley FPR
0.001	0.005	1	0.000	0.015	0.000	0.085
0.005	0.010	2	0.000	0.025	0.000	0.085
0.01	0.030	6	0.010	0.055	0.000	0.085
0.02	0.070	14	0.035	0.105	0.000	0.085
0.05	0.160	32	0.110	0.210	0.000	0.085
0.10	0.225	45	0.170	0.285	0.000	0.085
0.20	0.345	69	0.280	0.410	0.000	0.085

Table S2: Complete Parameter Sensitivity Results

Table 2: Grid size sweep. CSR data ($f = 0$) and clustered data ($f = 0.3$), 20 replicates each.

Grid size	FPR	TPR
16	0.15	1.00
32	0.05	1.00
64	0.05	1.00
128	0.10	1.00

Table S3: Hard Regime Detailed Results

Table S4: Realistic SMLM Conditions

Table S5: Runtime Scaling

Table 3: Shrinkage parameter sweep with mean condition numbers (20 replicates).

λ	FPR	TPR	Mean κ
0.01	0.10	1.00	600
0.05	0.00	1.00	619
0.10	0.15	1.00	600
0.20	0.05	1.00	529
0.50	0.10	1.00	413

Table 4: Number of mocks sweep (20 replicates).

M	FPR	TPR
5	0.25	1.00
10	0.10	1.00
20	0.10	1.00
50	0.15	1.00
100	0.10	1.00

Table 5: Blinking correction radius sweep (pipeline FPR). Mean reduction factor = (localizations before) / (localizations after merging).

r_{lat} (nm)	Pipeline FPR	TPR	Mean reduction
10	1.00	1.00	1.001
20	1.00	1.00	1.009
30	1.00	1.00	1.025
50	1.00	1.00	1.063
75	1.00	1.00	1.103

Table 6: Detection power in the hard regime (low cluster fraction and/or small cluster radius), 10 replicates each.

Configuration	Parameters	Multiscale TPR
Very sparse, standard radius	$f = 0.005, r = 75$ nm	0.20
Sparse, standard radius	$f = 0.01, r = 75$ nm	0.30
Low fraction, standard radius	$f = 0.02, r = 75$ nm	1.00
Moderate, very small radius	$f = 0.05, r = 15$ nm	1.00
Moderate, small radius	$f = 0.05, r = 30$ nm	1.00
Higher fraction, very small	$f = 0.10, r = 15$ nm	1.00
Low fraction, very small	$f = 0.02, r = 15$ nm	1.00

Table 7: Localization precision sweep ($f = 0.3$, 10 replicates).

σ_{xy} (nm)	Multiscale TPR
5	1.00
10	1.00
20	1.00
30	1.00
50	1.00
75	1.00
100	1.00

Table 8: Blinking rate sweep ($f = 0.3$, 10 replicates). Reduction factor = (raw localizations) / (merged molecules).

Blinks/molecule	Multiscale TPR	Reduction factor
1	1.00	1.004
3	1.00	1.015
5	1.00	1.026
10	1.00	1.051
20	1.00	1.104

Table 9: Frame count sweep ($f = 0.3$, 10 replicates).

Frames	Multiscale TPR
20	1.00
100	1.00
500	1.00
2000	1.00

Table 10: Full pipeline runtime at different dataset sizes. Includes blinking correction, null model generation (50 mocks), and multiscale testing. Single CPU core.

Molecules	Localizations	Wall clock (s)
500	2,514	29.7
1,000	5,013	31.6
2,000	10,047	35.1
5,000	25,122	51.3
10,000	50,014	247.4
20,000	99,999	3015

Supplementary Figures

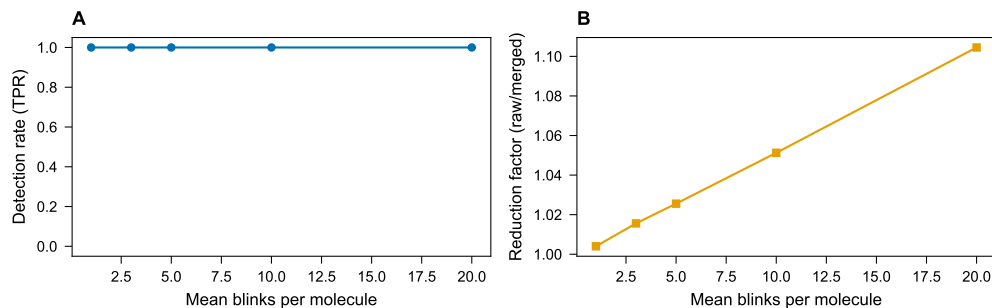


Figure 1: **Figure S1: Blinking correction performance.** Reduction factors (raw localizations / merged molecules) as a function of blinking rate (10 replicates). Even at 20 blinks per molecule, the correction reduces the localization count by only $\sim 10\%$, indicating conservative merging. Detection power (TPR) remains 1.0 across all blinking rates.

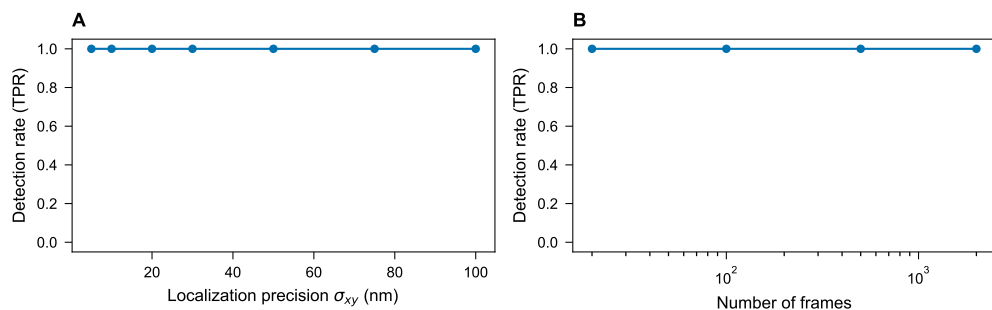


Figure 2: **Figure S2: Realistic SMLM conditions** (10 replicates). **(A)** Detection power vs. localization precision ($\sigma_{xy} = 5\text{--}100$ nm). TPR = 1.0 across all precisions. **(B)** Detection power vs. number of imaging frames (20–2000). TPR = 1.0 regardless of frame count.

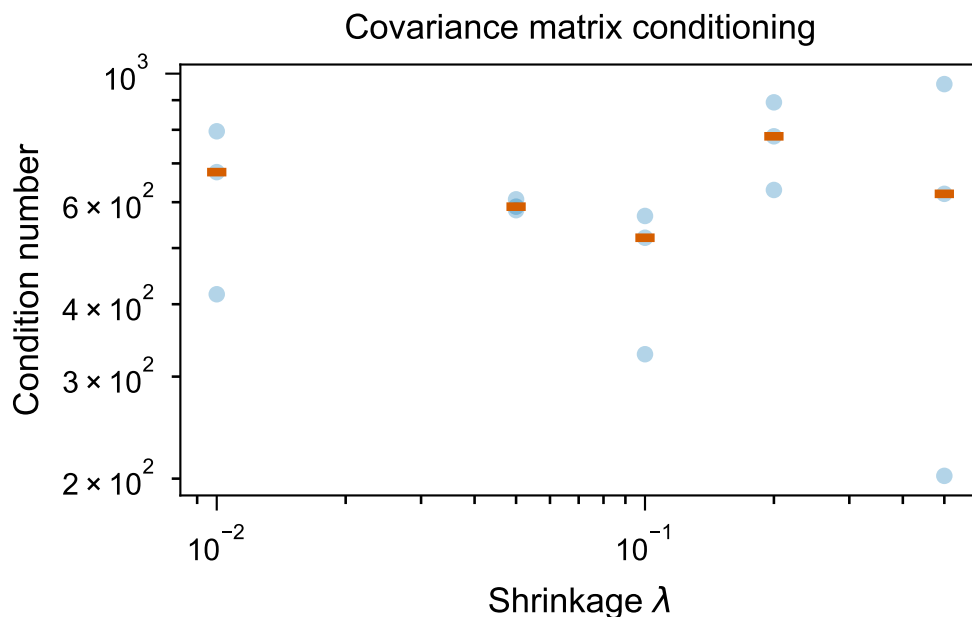


Figure 3: **Figure S3: Covariance matrix condition numbers** (20 replicates). Condition numbers of the regularized covariance matrix Σ_{reg} as a function of the shrinkage parameter λ . Mean condition numbers range from 413 ($\lambda = 0.5$) to 619 ($\lambda = 0.05$), ensuring stable matrix inversion across all settings.

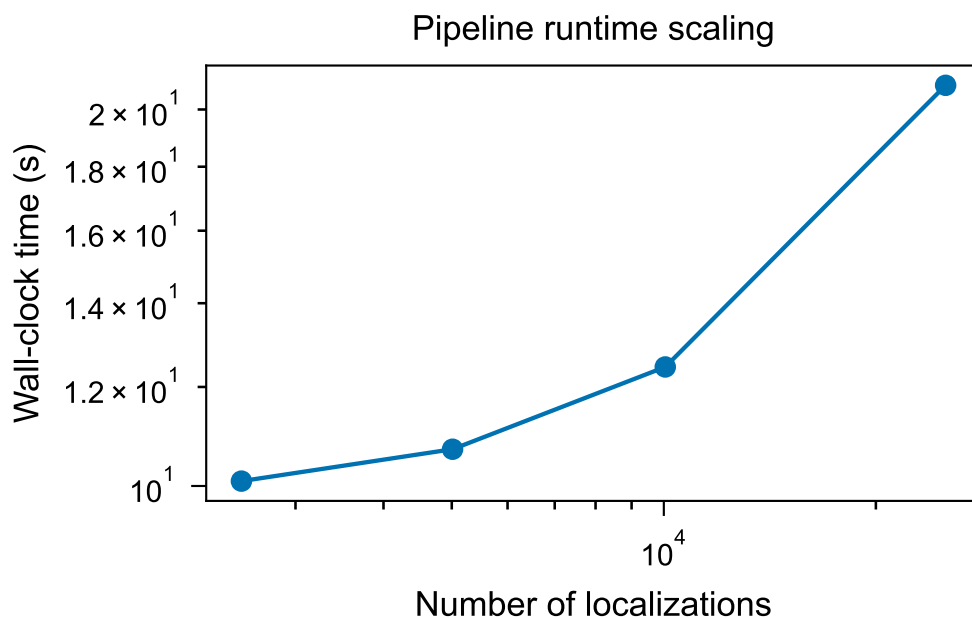


Figure 4: **Figure S4: Runtime scaling.** Wall-clock time for the full pipeline as a function of the number of molecules (500–20,000). Scaling is near-linear up to 5000 molecules (~ 51 s), with superlinear growth at larger sizes due to covariance estimation and mock generation costs.